

Chapter 1

Rings

1.1 Rings and Subrings

We now continue to generalize the familiar arithmetic operations of addition and multiplication. Just as with groups, results from the algebra of arithmetic will be extended to a host of non-numerical objects, e.g., matrices, functions, polynomials, etc.

Definition 1. A *ring* is a nonempty set, say R , with two binary operations (usually written as multiplication and addition)¹ that satisfies the following:

1. If $a, b \in R$ then $a + b \in R$. (closed under addition)
2. $a + (b + c) = (a + b) + c$ (associative addition)
3. $a + b = b + a$ (commutative addition)
4. There exists an 0_R such that $a + 0_R = 0_R + a = a$ (additive identity)
5. There exists an $-a \in R$ such that $a + (-a) = 0_R$ for every $a \in R$. (additive inverse)
6. If $a, b \in R$ then $ab \in R$. (closed under multiplication)
7. $a(bc) = (ab)c$ (associative multiplication)
8. $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ (distribution holds).

- Note that parts 1-5 say that a ring is an *abelian group*.²

Example 1. The set of integers, $\mathbb{Z}$, under addition and multiplication is a ring.

Example 2. $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ ($\mathbb{Z}$ modulo 4) is also a ring under addition and multiplication as in $\mathbb{Z}_n$ for any $n \in \mathbb{N}$.

Example 3. The set of 2 by 2 matrices with real numbers as entries,

$$\mathcal{M}_2(\mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}.$$

¹But of course the operations are not necessarily addition and multiplication. $+$ and $\times$; $\oplus$ and $\otimes$; are among the different notations that may be used in varying texts.

²Commutative groups are also called abelian in honor of Norwegian mathematician Niels Abel.

Closure, associativity, and distribution can be shown using the usual rules of matrix multiplication and addition. $\mathcal{M}_2(\mathbb{R})$ is commutative under addition (but not multiplication), the additive inverse is $\begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix}$, the additive identity is $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, and the multiplicative identity is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Proposition 1. *If a and b are two elements in R then $(-a)(-b) = ab$.*

Proof. By definition $a + (-a) = 0$. $(a + (-a))(-b) = 0$ implies $a(-b) + (-a)(-b) = 0$ by distribution. The additive inverses of $a(-b)$ is unique thus $a(-b) + (-a)(-b) = 0$ implies that $(-a)(-b) = ab$. $\square$

- In particular for rings like $\mathbb{Z}$ and $\mathbb{R}$, the product of two negatives is a positive. Note the importance of distribution.

Exercise 1. Show that $\mathbb{Z}_6$ is a ring with unity under addition and multiplication.

Exercise 2. Show that the set $\sqrt{2}\mathbb{Z} = \{x: x = m + \sqrt{2}n \text{ for some } m, n \in \mathbb{Z}\}$ is a ring.

Exercise 3. Let R and S be rings. Define addition and multiplication on the Cartesian product $R \times S$ by,

$$(r, s) + (r', s') = (r + r', s + s') \text{ and } (r, s)(r', s') = (rr', ss')$$

for $r, r' \in R$ and $s, s' \in S$. Prove that $R \times S$ is a ring.

Definition 2. A *commutative ring* is a ring R such that $ab = ba$ for all $a, b \in R$.

Definition 3. A ring with *identity*³ is a ring R with an element 1_R such that $a1_R = 1_Ra$ for all $a \in R$.

- It is important to check that a potential 1_R is both a *right identity*, $a1_R = a$; and a *left identity*, $1_Rb = b$.

Example 4. $\mathbb{C}$, $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{Z}$ are all commutative rings with identity under the usual addition and multiplication.

Example 5. $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ where $n \geq 2$ is a commutative ring with identity under the usual addition and multiplication (recall that n is called the *modulus*).

- To add $x + y \pmod n$ find $(x + y) \div n$ and find the remainder, say r , then $(x + y) \pmod n = r$. In the same way we can find $(xy) \pmod n$. For example, $(3 \cdot 7) \pmod 8 = 5$ since $2 \cdot 8 + 5 = 3 \cdot 7$ (divide by 8 and take the remainder).

Proof. Since $r < n$, $\mathbb{Z}_n$ will be closed under addition and multiplication. Parts 2, 3, 4, and 7 hold because of the normal rules of arithmetic. Which also give multiplicative commutativity. 0 and 1 are the additive and multiplicative identities respectively.

Which only leaves distribution, part 8, and additive inverse, part 5. Let $a \in \mathbb{Z}_n$ then $a < n$ so then $n - a \in \mathbb{Z}_n$, and $a + (n - a) = 0$. For distribution, let $a, b, c \in \mathbb{Z}_n$. $a(b + c) \pmod n = (ab + ac) \pmod n = ab \pmod n + ac \pmod n$ (this was shown last semester). $\square$

Example 6. $E = \{x: x = 2n \text{ for some } n \in \mathbb{Z}\}$, the set of even integers is a commutative ring. However it does *not* have identity.

³The term *unity* is often used in place identity. A ring with an identity element is said to “have unity”.

- Keep in mind that a ring is *not* a group under multiplication. Elements in a ring do not necessarily have a multiplicative inverse, and rings do not necessarily have a multiplicative identity. Keep examples of rings in mind which do not have these properties.

Exercise 4. Prove that a ring that is cyclic⁴ under addition and commutative (under multiplication).

Theorem 1. *If a ring has an identity, it is unique.*

Proof. Suppose to the contrary that a ring R has two identity elements 1 and $1'$. $a1 = a$ for all $a \in R$ including $1' \in R$. Thus $1' \cdot 1 = 1'$ and $1 \cdot 1' = 1$. By the definition of a ring $1' = 1 \cdot 1' = 1' \cdot 1 = 1$. Therefore $1' = 1$, and the multiplicative identity is unique. $\square$

Exercise 5. Let F be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. Define addition and multiplication as

$$(f + g)(x) = f(x) + g(x) \text{ and } (fg)(x) = f(x)g(x).$$

Show that F is a commutative ring with identity.

Exercise 6. If a ring element has a multiplicative inverse, it is unique.

1.2 Subrings

Definition 4. Let R be a ring and S be a subset of R .⁵ If S is a ring under the same operations as R , then S is a *subring*.

Example 7. $\mathbb{Q}$ is a subring of $\mathbb{R}$, and $\mathbb{Z}$ is a subring of $\mathbb{Q}$ and $\mathbb{R}$.

Example 8. The set of even integers is a subring of $\mathbb{Z}$, but the set of odd integers is not since it is not closed under addition.

The following two theorems provide shortcuts when trying to show that a subset is a subring. Particularly, The second theorem can save quite a bit of work.

Theorem 2. *Suppose that R is a ring and $S \subseteq R$ such that the following hold:*

1. *If $a, b \in S$ then $a + b \in S$ (closed under addition)*
2. *If $a, b \in S$ then $ab \in S$ (closed under multiplication)*
3. $0_R \in S$.
4. *If $a \in S$ then there is an $-a \in S$ such that $a + (-a) = 0_R$.*

Then S is a subring of R .

Proof. Parts 2, 3, 7, and 8 of the Ring definition hold for all elements in R since it is a ring. So they hold for all elements in S . The remaining requirements 1-4 in the theorem satisfy the remaining requirements. $\square$

- Rings and thus subrings must be nonempty. A requirement which is covered by 3. Since $0_R \in S \Rightarrow S \neq \emptyset$.

⁴Recall that a group G is *cyclic* if for every $x \in G$ there is some $g \in G$ such that $x = g + g + \cdots + g$. In which case we write $G = \langle g \rangle$ and say that G is *generated* by g , e.g., $\mathbb{Z}_3 = \langle 1 \rangle = \langle 2 \rangle$ since $2 + 2 = 1$, $2 = 2$, and $2 + 2 + 2 = 0 \pmod{3}$.

⁵Notated: $S \subseteq R$.

Theorem 3. A nonempty subset S of a ring R is a subring if whenever $a, b \in S$:

1. $ab \in S$ (closed under multiplication)
2. $a - b \in S$ (closed under subtraction)

Proof. If $a - b \in S$ for all $a, b \in S$ then S is a subgroup of R (subgroup test), and since R is additively commutative it follows that S is an abelian subgroup of R under addition (this also means that $0_R \in S$). S is multiplicatively and additively associative; and S is distributive over addition because these properties hold in R . $\square$

Exercise 7. Show that the set $3\mathbb{Z} = \{x : x = 3n \text{ for some } n \in \mathbb{Z}\}$ is a subring of $\mathbb{Z}$.

Exercise 8. Show that the set $k\mathbb{Z} = \{x : x = kn \text{ for some } n \in \mathbb{Z}\}$ for any integer k is a subring of $\mathbb{Z}$.

Exercise 9. Prove that the intersection of any two subrings of a ring R is also a subring of R .

1.3 Integral Domains and Fields

For $\mathbb{Z}_n$ it is easy to see that $ab = 0$ does *not* imply that a or b is 0. For example, in $\mathbb{Z}_6$ $2 \cdot 3 \bmod 6 = 6 \bmod 6 = 0$. Of course many rings (indeed the most familiar rings) do have this property. In $\mathbb{R}$, if $x \cdot y = 0$ then $x = 0$ or $y = 0$.

Definition 5. An *integral domain* is a commutative ring R with identity such that $1_R \neq 0_R$ ⁶ and if $ab = 0_R$ then $a = 0_R$ or $b = 0_R$ for any $a, b \in R$.⁷

Example 9. $\mathbb{Z}$ is an integral domain.

Let $a, b \in \mathbb{Z}$. Suppose that $b \neq 0$ (if $b = 0$ then the condition is met) then $ab = 0 \Rightarrow a = 0/b = 0$.

Example 10. The ring $\mathbb{Z}[x]$, the polynomials with integer coefficients, is an integral domain.

Exercise 10. Show that $\mathbb{Z}_6$ is *not* an integral domain.

Exercise 11. Prove that $\mathbb{Z}_n$ is not an integral domain if n is not prime.

Hint If n is not prime then $n = mq$ for two integers $1 < m, q < n$. Looking at the example seen in exercise might help.

Exercise 12. Prove that $\mathbb{Z}_p$ is an integral domain if p is prime.

Exercise 13. Prove that $\mathbb{Q}$ is an integral domain.

Exercise 14. Prove that the ring from exercise 5, F , is not an integral domain.

Hint Here are two *indicator functions*

$$\mathbf{1}_{[0, \infty)}(x) = \begin{cases} 1 & \text{if } x \in [0, \infty) \\ 0 & \text{if } x \notin [0, \infty) \end{cases} \text{ and } \mathbf{1}_{(-\infty, 0]}(x) = \begin{cases} 1 & \text{if } x \in (-\infty, 0] \\ 0 & \text{if } x \notin (-\infty, 0] \end{cases}$$

The indicator function is a function that has an output of either 1 or 0. For example, $\mathbf{1}_{[0, \infty)}(2) = 1$ and $\mathbf{1}_{[0, \infty)}(-2) = 0$. Now consider the functions $f(x) = x\mathbf{1}_{[0, \infty)}$ and $g(x) = x\mathbf{1}_{(-\infty, 0]}$.

⁶The condition $1_R \neq 0_R$ is necessary so that the ring $\{0\}$ is not an integral domain.

⁷If a, b are nonzero elements in a ring and $ab = 0$ then a and b are called *zero divisors*. Thus an integral domain is a commutative ring with identity that has no zero divisors.

Example 11. The ring of 2×2 matrices with real entries, $M_{2 \times 2}(\mathbb{R})$, is not an integral domain.

Proof. This ring has zero $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ so the matrices $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ are nonzero and $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. $\square$

When two nonzero elements in a ring have a product that is 0, we call the elements *zero divisors*. Having no such elements allows for cancellation by the following theorem.

Theorem 4. Let R be an integral domain and let $a, b, c \in R$ where $c \neq 0$. If $ac = bc$, then $a = b$.

Proof. Since rings have distribution, $ac = bc \Rightarrow ac - bc = 0 \Rightarrow (a - b)c = 0$. Because R is an integral domain $a - b = 0$ which implies that $a = b$. $\square$

Note that we could not prove theorem 4 using division, i.e., $\frac{ac}{c} = \frac{bc}{c}$, because we cannot assume that $c^{-1} \in R$. For example, $3x = 3y \Rightarrow x = y$ in $\mathbb{Z}$ but $\frac{1}{3} \notin \mathbb{Z}$. While $\mathbb{Z}$ is an integral domain, it has no element x such that $3x = 1$. That is, 3 (and all other numbers except 1) does not have a multiplicative inverse. All the elements in $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ (except 0) do have multiplicative inverses. Thus $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are all examples of the following definition.

Exercise 15. Let a and b be two nonzero elements in a ring R . Prove that in general, $ax = b$ does not have a unique solution.

Hint It should be understood that x must be in R .

Definition 6. A *field* is a commutative ring R with identity $1_R \neq 0_R$ ⁸ if for every $a \neq 0_R$ in R , there exists an $a^{-1} \in R$ such that $aa^{-1} = 1_R$.

- Sets $\supseteq$ Groups $\supseteq$ Rings $\supseteq$ Commutative Rings with identity $\supseteq$ Integral Domains $\supseteq$ Fields

Example 12. $\mathbb{R}$ and $\mathbb{Q}$ are fields.

Example 13. The set of *constructable numbers*, numbers which can be constructed with a straight edge and compass, is a field.

Exercise 16. Prove that $\mathbb{C}$ is a field.

Theorem 5. $\mathbb{Z}_p$ is a field if p is a prime.

Proof. This is left as an exercise. $\square$

Theorem 6. Cancellation. Let a, b, c be elements in an integral domain. Suppose that $a \neq 0$. Then $ab = ac \Rightarrow b = c$.

Proof. Let $a \neq 0$ for elements a, b, c in an integral domain, R . $ab = ac \Rightarrow a(b - c) = 0_R$. Because R is an integral domain either $a = 0$ or $b - c = 0$. By assumption, $a \neq 0$ so then $b - c = 0 \Rightarrow b = c$. $\square$

Theorem 7. A field is an integral domain.⁹

Proof. This is left as an exercise. $\square$

⁸Once again, this assumption is needed to that $\{0\}$ is excluded from this class.

⁹Often a *field* is defined as an integral domain in which every nonzero element has an multiplicative inverse.

Theorem 8. *A finite integral domain is a field.*

Proof. Let R be a finite integral domain with identity. Let $a \in R$ and $a \neq 0_R$. All we need to show is that a has an inverse. If $a = 1_R$ then $a^{-1} = 1_R$. Now suppose that $a \neq 1_R$ (if no such exists then we are done). Since R is finite there are two integers such that $i < j$ and $a^i = a^j$ (otherwise R would be infinite). By cancellation (above theorem) we have that $a^{j-i} = 1_R$. Because $a \neq 1_R$ we know that $j - i > 1$. Thus $a^{j-i-1}(a) = 1_R$ as needed. $\square$

Exercise 17. Prove that a field is an integral domain.